Estimating the Causal Treatment Effect of Job Training Programs on Participant Earnings

Evidence from the U.S.

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Github: https://github.com/lefie1231/CausalInference_ML

Content

Look at Data & Structural Model

Extract Treatment Effects

Does participation in a job training program increase real earnings for disadvantaged people?

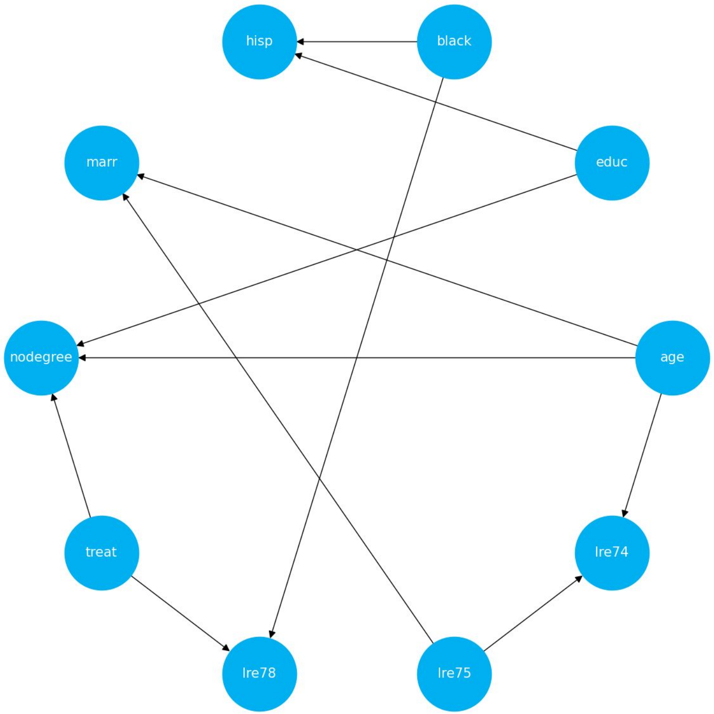
Inspect Heterogeneity

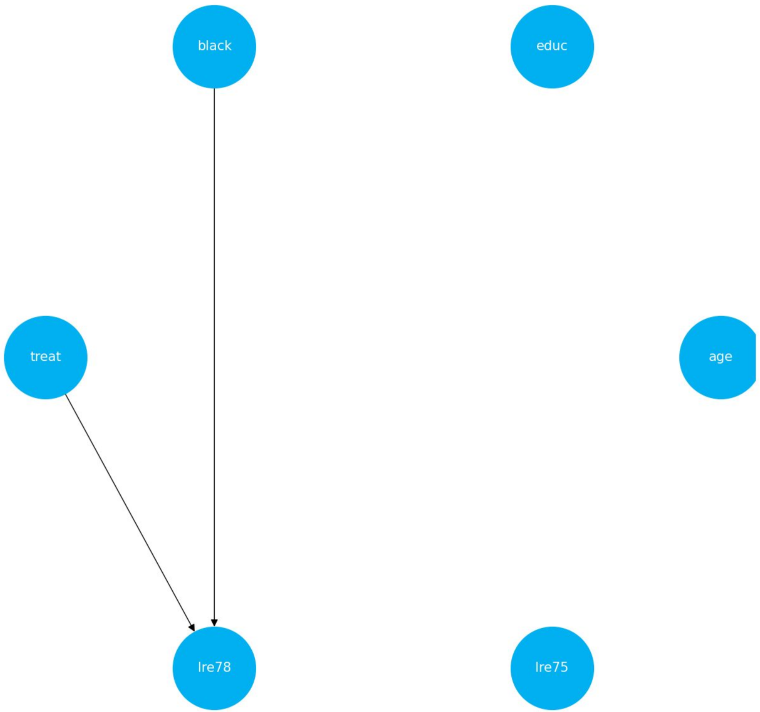
Which group benefits the most?

Data

	treat	age	educ	black	hisp	marr	nodegree	re74	re75	re78
count	445	445	445	445	445	445	445	445	445	445
mean	0,42	25,37	10,20	0,83	0,09	0,17	0,78	2.102,27	1.377,14	5.300,76
std	0,49	7,10	1,79	0,37	0,28	0,37	0,41	5.363,58	3.150,96	6.631,50
min	0	17	3	0	0	0	0	0	0	0
25%	0	20	9	1	0	0	1	0	0	0
50%	0	24	10	1	0	0	1	0	0	3.701,81
75%	1	28	11	1	0	0	1	824,39	1.220,84	8.124,71
max	1	55	16	1	1	1	1	39.570,68	25.142,24	60.307,93

Causal Structural Model

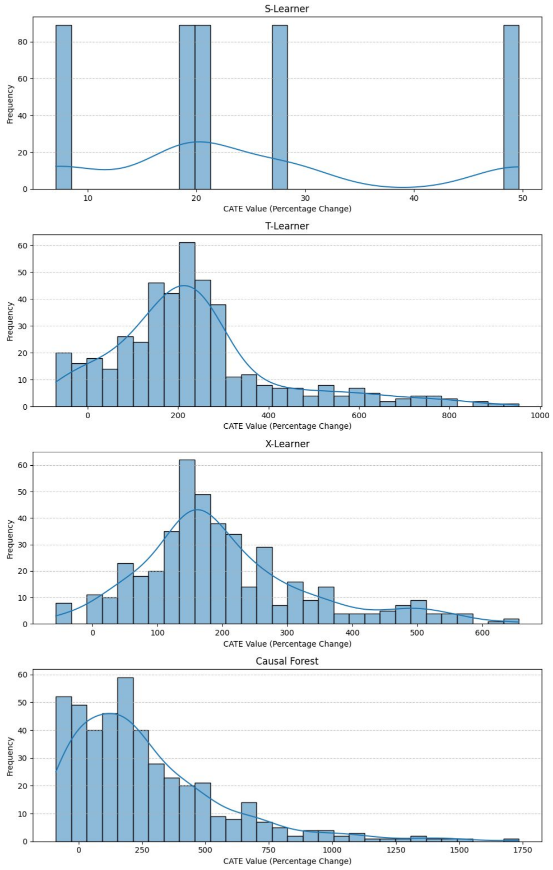




Model Selection

Regression Models:	Mean squared error	Adjusted R-squared
Linear OLS	13.6676	-0.0836
Linear Ridge	13.6407	-0.0815
Linear LAVA	13.6676	-0.0836
Flexible OLS	13.8815	-0.1278
Flexible Ridge	13.5649	-0.1021
Flexible LAVA	13.8815	-0.1278

Linear model: $$\log(re_{78}) = \beta_0 + \beta_1 treat + \beta_2 black + \beta_3 educ + \beta_4 age + \beta_5 \log(re_{75}) + \epsilon$$



Forest Models

R-loss:

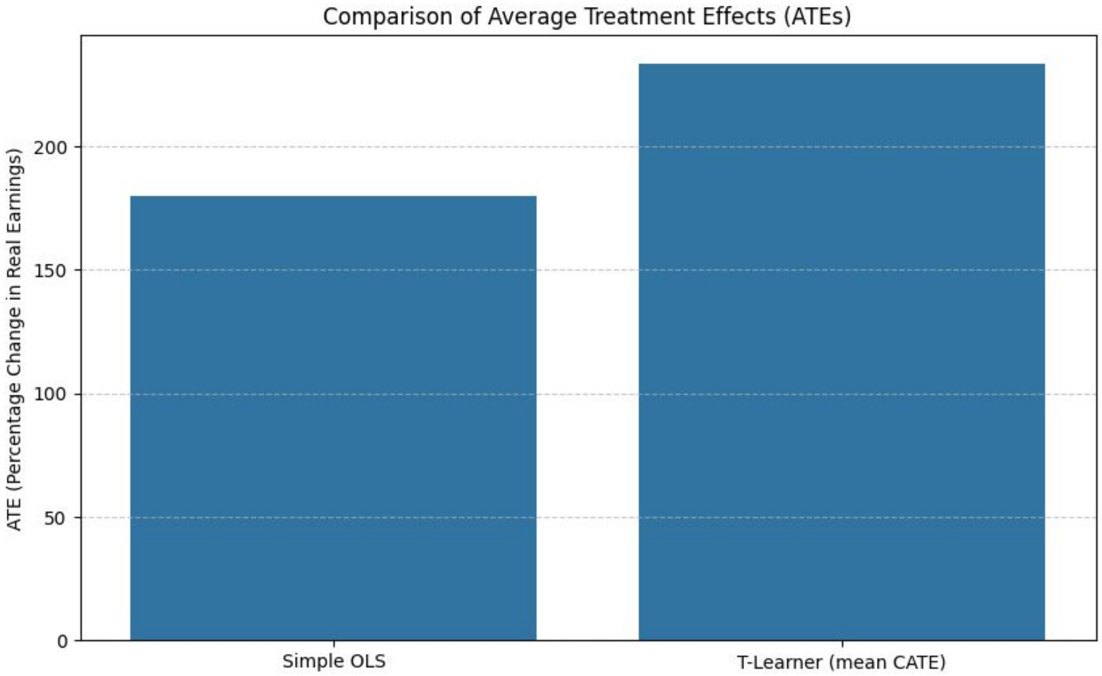
-0.0113

0.0037

0.0084

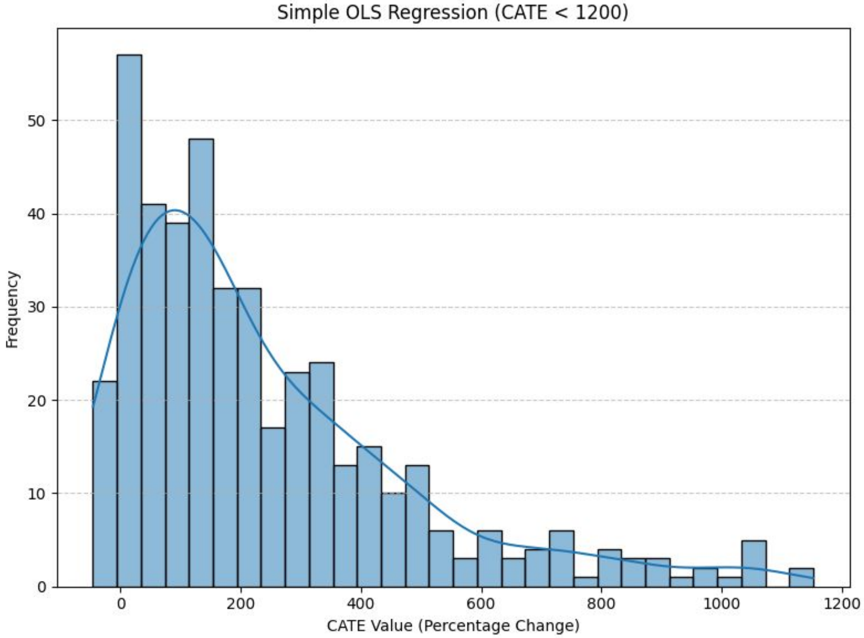
0.0285

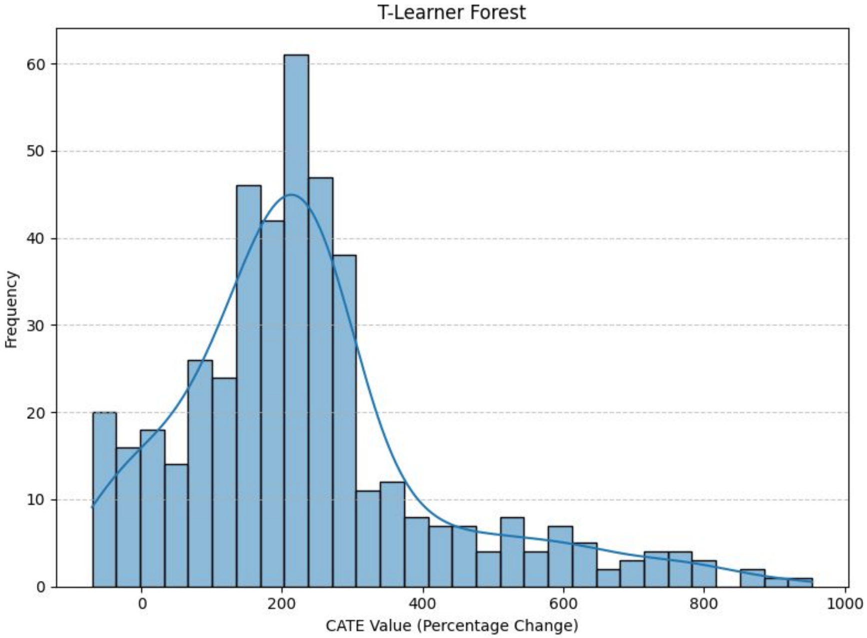
Extracting Treatment Effects - ATE



179.93%	233.29%
(+ 5235.90$ for bottom quintile)	(+ 6788.68$ for bottom quintile)

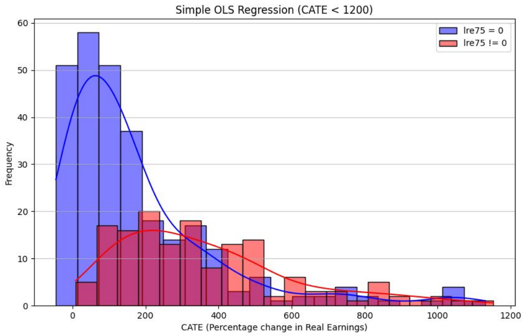
Extracting Treatment Effects - CATE



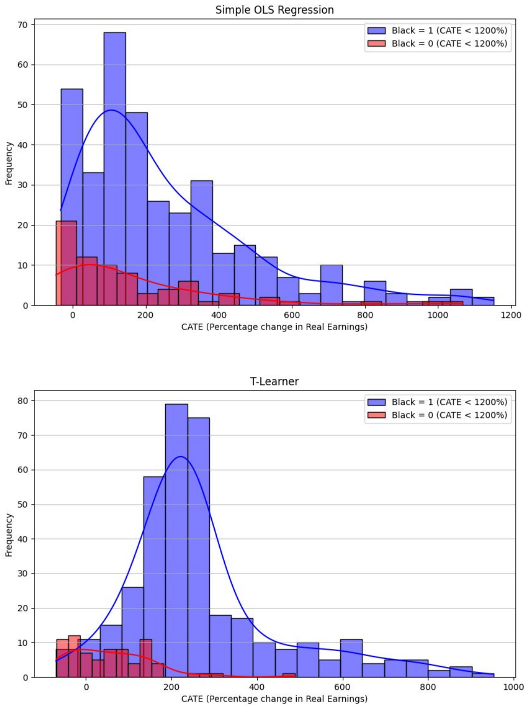


Inspecting Heterogeneity

By Pre-Treatment Income

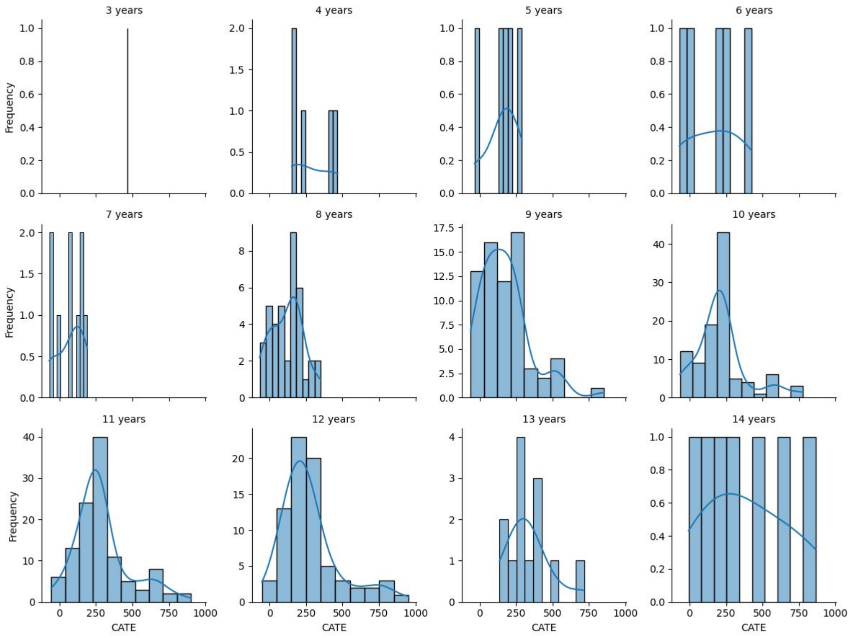


By Race

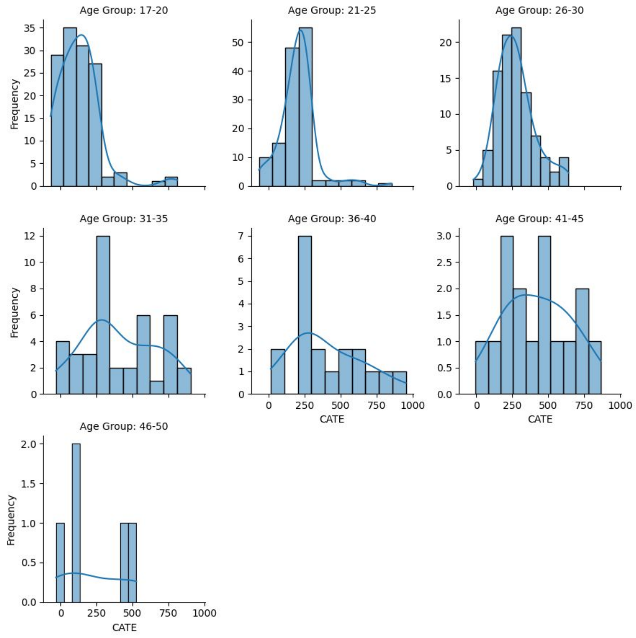


Inspecting Heterogeneity

By Education (T-Learner CATE < 1200%)



By Age (T-Learner, CATE < 1200%)



Conclusion

Does participation in a job training program increase real earnings for disadvantaged people?

- Yes, average treatment effect is positive
- Some participants face earnings decreases

Which group benefits the most?

- Those with non-zero pre-treatment earnings
- Black participants
- Those with higher education (9 years +)
- Those that are older than 26

Thank you for your attention!